



<https://www.docs.isalos.novamechanics.com/>

The objective of this practical session is to investigate how the physicochemical properties of per- and polyfluoroalkyl substances (PFAS) influence their environmental fate and potential human exposure. We will apply a structured modelling workflow that integrates environmental fate simulations, statistical analysis, and machine-learning modelling using three computational tools: SimpleBox4Planet (<https://www.enalosccloud.novamechanics.com/proplanet/simplebox4planet/>) and Isalos Analytics Platform (<https://enalosccloud.novamechanics.com/novamechanicssystem/userregistration/>).

In addition to evaluating the influence of key physicochemical properties, the practical will aim to identify combinations of property values and levels that minimize environmental mass, thereby supporting the design of safer alternative PFAS within a Safe-and-Sustainable-by-Design (SSbD) framework. Furthermore, a machine-learning surrogate model will be developed to predict environmental mass directly, reducing the need to repeatedly run the underlying mechanistic mass-balance and differential-equation models and thus lowering the computational cost of screening.

Please, follow the steps presented in the following link to download and install Isalos Analytics on your local PCs: <https://www.docs.isalos.novamechanics.com/installation.html>.

Isalos version used: 2.0.2

Step 1: Import data

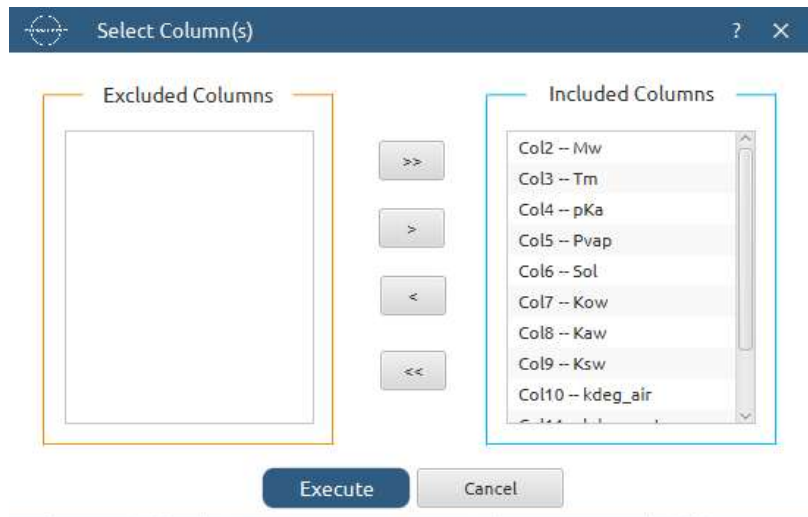
Manually insert data as shown in the table below into Isalos on the input spreadsheet (left).

Mw	Tm	pKa	Pvap	Sol	Kow	Kaw	Ksw	kdeg_air	kdeg_water	kdeg_sed	kdeg_soil
200.0	40.0	-3.0	1.0E-9	10.0	100.0	1.0E-8	0.1	1.0E-7	1.0E-8	1.0E-11	1.0E-10
800.0	350.0	4.0	0.01	10000.0	1000000.0	0.001	100.0	1.0E-6	1.0E-7	1.0E-9	1.0E-8

	Col1	Col2 (D)	Col3 (D)	Col4 (D)	Col5 (D)	Col6 (D)	Col7 (D)	Col8 (D)	Col9 (D)	Col10 (D)	Col11 (D)	Col12 (D)	Col13 (D)
User Header	User Row ID	Mw	Tm	pKa	Pvap	Sol	Kow	Kaw	Ksw	kdeg_air	kdeg_water	kdeg_sed	kdeg_soil
1		200.0	40.0	-3.0	1.0E-9	10.0	100.0	1.0E-8	0.1	1.0E-7	1.0E-8	1.0E-11	1.0E-10
2		800.0	350.0	4.0	0.01	10000.0	1000000.0	0.001	100.0	1.0E-6	1.0E-7	1.0E-9	1.0E-8

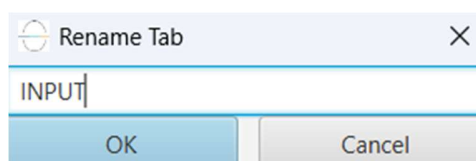
Step 2: Manipulate data

Select all columns by navigating *Data Transformation > Data Manipulation > Select Column(s)* and click on 'Execute' button.



User Header	Col1	Col2 (D)	Col3 (D)	Col4 (D)	Col5 (D)	Col6 (D)	Col7 (D)	Col8 (D)
User Row ID	Mw	Tm	pKa	Pvap	Sol	Kow	Kaw	Ksw
1	2000	40.0	-3.0	1.0E-9	10.0	100	0E-7	0E-7
2	8000	3500	4.0	0.01	100000	1000000	0.001	
3								
4								
5								
6								
7								
8								
9								
10								
11								
12								
13								
14								

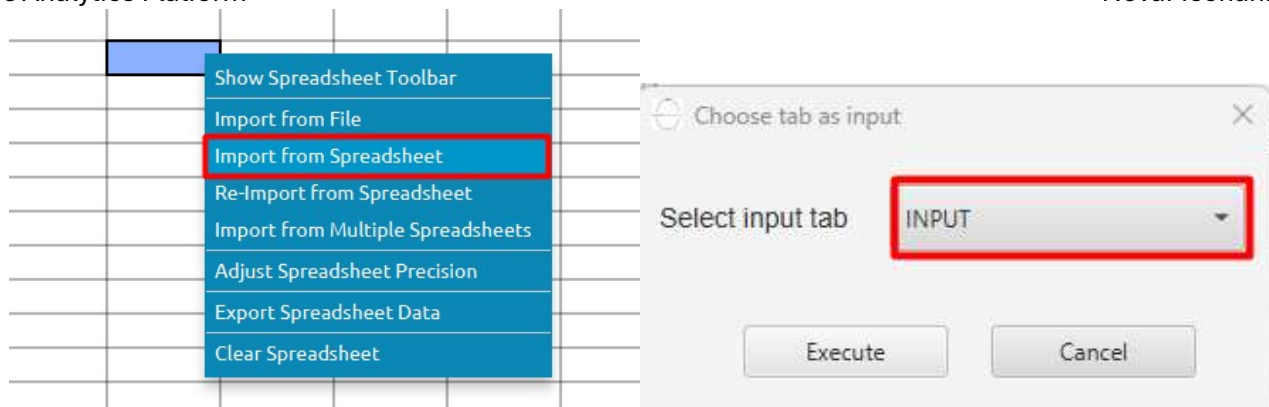
Rename the sheet to 'INPUT' by right-clicking on the node and selecting 'Rename'.



Step 3: Create new sheet for Plackett Burman

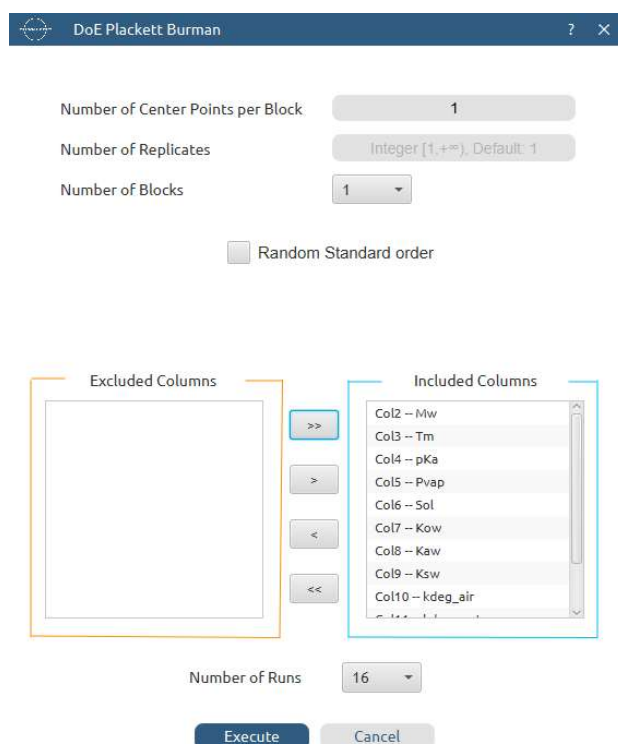
Create a new sheet and name it as 'PLACKETT BURMAN'.

Right-click on the input (left) spreadsheet and select 'Import from Spreadsheet'. Then select 'INPUT'.



Step 4: Plackett Burman

Design Plackett Burman by selecting from the tabs *DOE > Screening > Plackett Burman*. Once the configuration window appears, make the selections as indicated below and proceed by clicking the 'Execute' button.



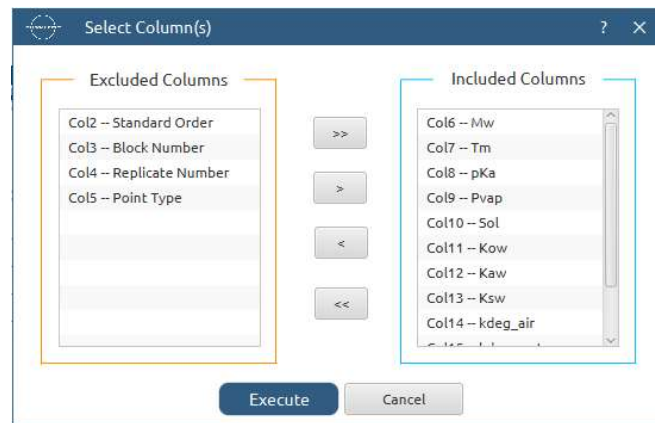
Hint:

The analysis began with a screening design, specifically a Plackett–Burman design, to identify the most influential factors affecting the environmental fate response and to provide an initial indication of whether the system could be described using a linear model. Since the Plackett–Burman design is primarily intended for screening purposes, it can only estimate main effects and does not capture interaction or quadratic terms.

Step 5: PREPROCESS: Import & Select

Create a new sheet and name it as 'PREPROCESS'. Import data from the output of the 'PLACKETT BURMAN' sheet.

Then select from tabs *Data Transformation > Data Manipulation > Select Column(s)*. Ensure to replicate the selection of columns as indicated below, and then click the 'Execute' button.



Step 6: Export the output as a CSV file

Export the output to a CSV file by right-clicking on the output spreadsheet (on the right) and selecting 'Export Spreadsheet Data.' Choose 'CSV' as the file extension and click the 'Execute' button.

Name the file 'ScreeningDesign' and save it. Subsequently, open the file, delete the first column, and save the file again.

10.0	1000000.0	0E-7	100.0	0.0000010	1E-
10000.0	100.0	0.001	100.0	0.0000010	1E-
10.0	1000			0.0000010	0E-
10000.0	1000			1E-7	0E-
10000.0	1000			1E-7	0E-
10000.0	1000			1E-7	1E-
10000.0	100.0	0E-7	0.1	0.0000010	0E-

Compute environmental mass using SimpleBox4Planet

Step 7: Access Sensitivity Exploration

Open the following URL link:

<https://www.enalosccloud.novamechanics.com/proplanet/simplebox4planet/> and navigate to the bottom of the page. Select 'Sensitivity Exploration' button.

Step 8: Upload Input File

When a new window appears, click on 'Upload CSV' button and select the file 'ScreeningDesign.csv' which created in the previous step.

Step 9: Execute & Download Results

After the data is successfully loaded, click on the 'Execute' button to generate the total mass produced (in kg).

Finally, click on the 'Download Results (Excel)' button to download the results, naming the file 'ScreeningDesignResults.xlsx'.

Sensitivity Exploration

Buttons: Add Row, Remove Selected, Upload CSV, Download CSV Template, Clear, Execute, Download Results (Excel)

Input scenarios (Mw → kdeg(soil)). Add rows manually or upload CSV.

Row	Mw [g mol ⁻¹]	Tm [°C]	pKa	Pvap [Pa]	Sol25 [mg L ⁻¹]	Kaw	Kow	Ksw	kdeg(air) [s ⁻¹]	kdeg(water) [s ⁻¹]	kdeg(sed) [s ⁻¹]	kdeg(soil) [s ⁻¹]	Total (kg)
☐	200	40	-3	0.01	10	100	0.001	100	1e-7	1e-7	1e-11	1e-8	4.19432e+07
☐	200	40	4	1e-9	10	1000000	0.001	0.1	0.000001	1e-8	1e-9	1e-8	3.83110e+07
☐	200	350	-3	1e-9	10000	1000000	1e-8	100	1e-7	1e-7	1e-9	1e-8	4.52409e+07
☐	800	40	-3	0.01	10000	100	0.001	0.1	0.000001	1e-7	1e-9	1e-8	1.73039e+07
☐	200	40	4	0.01	10	1000000	1e-8	100	0.000001	1e-7	1e-9	1e-10	3.17755e+09
☐	200	350	4	1e-9	10000	100	0.001	100	0.000001	1e-7	1e-11	1e-10	2.84037e+08
☐	800	350	-3	0.01	10	1000000	0.001	100	0.000001	1e-8	1e-11	1e-10	3.19557e+09
☐	800	40	4	1e-9	10000	1000000	0.001	100	1e-7	1e-8	1e-11	1e-8	4.73348e+07

Execution completed for 17 rows.

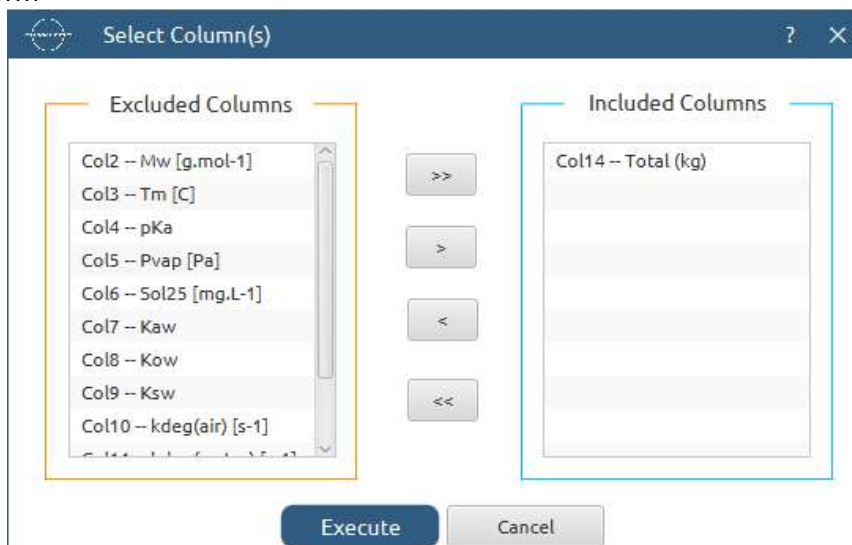
Statistical Analysis using Isalos

Step 10: Create Sheet & Import Results

Create a new sheet in Isalos Analytics and name it as 'RESULTS-SCREENING'. Right-click on the input (left) spreadsheet and select 'Import from File' to import the ScreeningDesignResults.xlsx created in the previous step.

Step 11: Select Output Column

Then navigate to *Data Transformation > Data Manipulation > Select Column(s)* and select only the column corresponding to the generated output, as shown below.



Step 12: PREPROCESS-v2: Merge Data

Create a new sheet and name it as 'PREPROCESS-v2'. Right-click on the input (left) spreadsheet and select 'Import from Multiple Spreadsheets'. Select the 'PREPROCESS' as the Left Spreadsheet and 'RESULTS-SCREENING' as the Right Spreadsheet, then click on 'Execute' button.

	Col1	Col2	Col3	Col4	Col5	Col6
User Header	User Row ID					
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						

Show Spreadsheet Toolbar
 Import from File
 Import from Spreadsheet
 Import from Multiple Spreadsheets
 Adjust Spreadsheet Precision
 Export Spreadsheet Data
 Clear Spreadsheet

Join Configuration Steps

Step 1: PREPROCESS RESULTS-SCREEN

Join Type

☒ Concatenation ☐ Left Join ☐ Right Join ☐ Inner Join ☐ Full Outer Join

Left Spreadsheet

PREPROCESS

Right Spreadsheet

RESULTS-SCREENING

Join Column

☒ Common header name ☐ Different header names

Execute Cancel

Step 13: PREPROCESS-v2: Select Columns

Navigate to *Data Transformation > Data Manipulation > Select Column(s)* and select the columns shown below.

Select Column(s)

Excluded Columns

Col14 -- User Row ID_RESULTS

Included Columns

Col2 -- Mw
Col3 -- Tm
Col4 -- pKa
Col5 -- Pvap
Col6 -- Sol
Col7 -- Kow
Col8 -- Kaw
Col9 -- Ksw
Col10 -- kdeg_air

Execute Cancel

Step 14: PARETO

Create a new sheet and name it as 'PARETO'. Right-click on the input (left) spreadsheet and select 'Import from Spreadsheet' and select the 'PREPROCESS-v2'. Navigate to *DOE > Post DoE Analysis > Pareto Analysis* and select the configuration as shown below.

Pareto Analysis

?

×

Dependent Variable

Col14 ~ Total (kg)

Analysis Type

Main Effects

Level Of Significance

0.05

Excluded Columns

>

<

Factors

Col2 ~ Mw

Col3 ~ Tm

Col4 ~ pKa

Col5 ~ Pvap

Covariates

>

<

DOE type:

☒ Factorial / Screening

☐ Response Surface

☒ Include Center Points

Execute

Cancel

Output:

Pareto Chart for Standardized Effects: Total (kg)

File

View

Pareto Chart for Standardized Effects: Total (kg)

(a = 0.05)

Term

kdeg_soil

Kow

Pvap

Curvature

Ksw

Sol

kdeg_air

kdeg_water

pKa

Tm

kdeg_sed

Kaw

Mw

Effect

0

5

10

15

20

25

30

35

40

45.679

3.18

Customization

Background:

White

Bar Color:

#f3622d

Bar Width: 1.00

8

Observation:

The analysis indicates that kdeg(soil), Kow and Pvp effects exceed the statistical significance threshold and are therefore identified as the dominant factors influencing the environmental fate response. All remaining variables fall below the threshold and are considered non-significant within the explored design space and thus are excluded from further analysis.

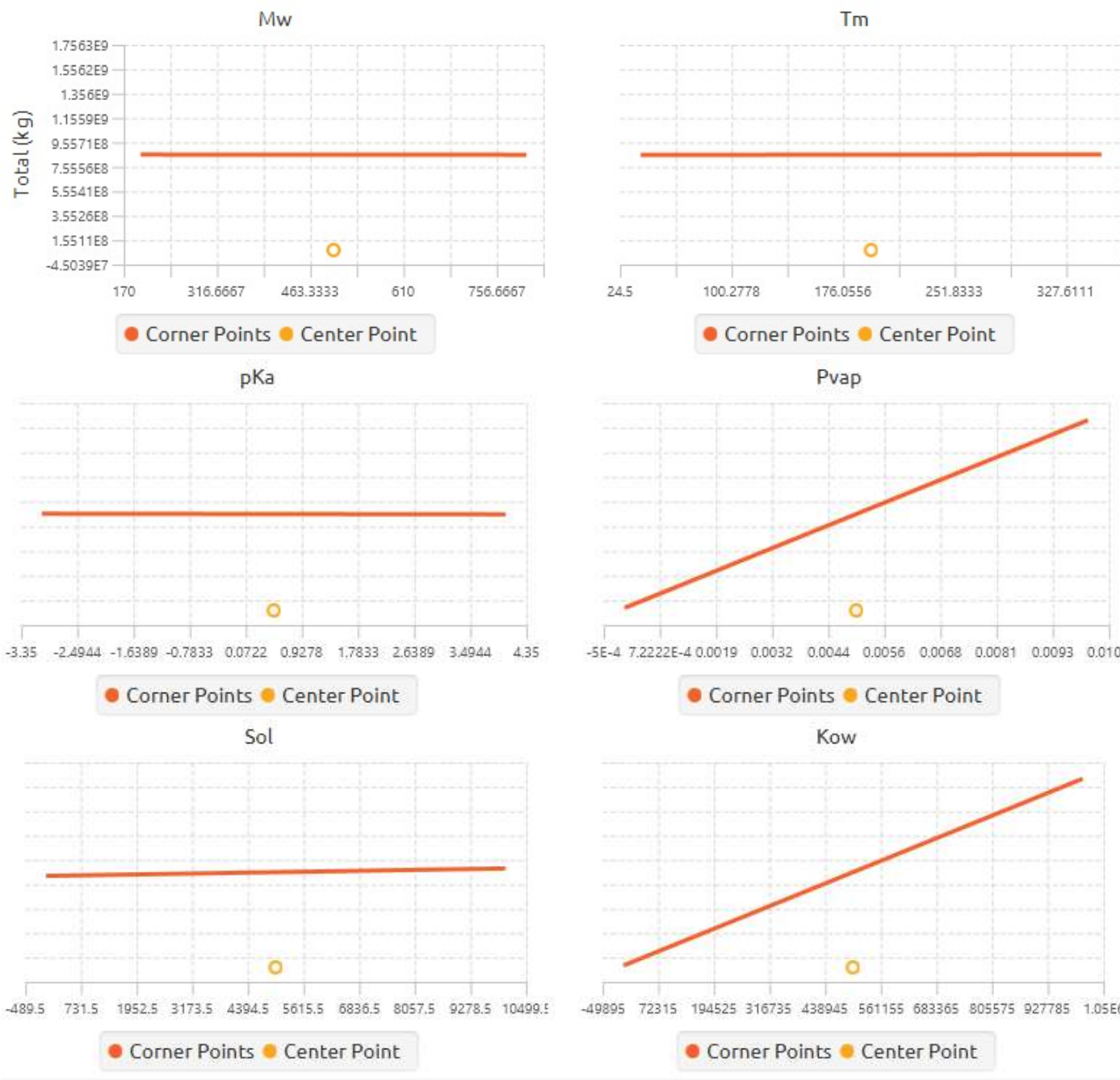
Step 15: Factorial-Plots

Create a new sheet and name it as 'FACTORIAL-PLOTS'. Right-click on the input (left) spreadsheet and select 'Import from Spreadsheet' and select the 'PREPROCESS-v2'.

Navigate to *DOE > Post DoE Analysis > Factorial Plot Analysis* and select the configuration as shown below.

The screenshot shows the 'Factorial Plot Analysis' dialog box. At the top, the title bar reads 'Factorial Plot Analysis'. Below the title bar, there are two dropdown menus: 'Dependent Variable' set to 'Col14 ~ Total (kg)' and 'Analysis Type' set to 'Main Effects'. In the center, there are three main sections: 'Excluded Columns' (an empty box with an orange border), 'Factors' (a list box containing 'Col2 ~ Mw', 'Col3 ~ Tm', 'Col4 ~ pKa', and 'Col5 ~ Pvp'), and 'Covariates' (an empty box with a blue border). Between 'Excluded Columns' and 'Factors' are two arrow buttons (> and <). Below the 'Factors' list box is a button labeled 'Specify Factor Values'. At the bottom, there are two radio buttons for 'DOE type': 'Factorial / Screening' (selected) and 'Response Surface'. Below these is a checked checkbox labeled 'Include Center Points'. At the very bottom are two buttons: 'Execute' and 'Cancel'.

Output:





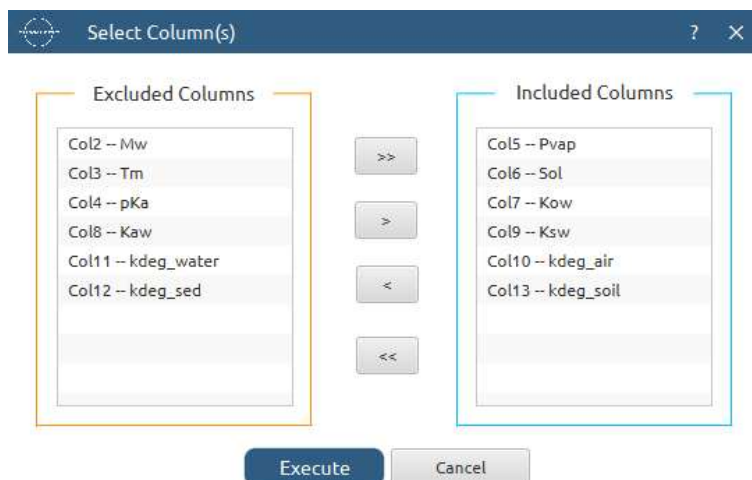
Observation:

The factorial plot analysis indicates that a linear model is not sufficient to describe the system behavior. For all examined factors, the center points do not lie on the straight line connecting the low and high factor levels, indicating the presence of curvature in the response. This deviation suggests that the relationship between the physicochemical properties and the environmental fate response is non-linear, thereby supporting the need for an RSM design to adequately capture these curvature effects.

Step 16: PREPROCESS- v3: Import & Select

Create a new sheet and name it as 'PREPROCESS-v3'. Import data from the 'INPUT' sheet.

Navigate to *Data Transformation > Data Manipulation > Select Column(s)* and select columns according to the specified selections below.

**Hint:**

Since the significant factors had already been identified (step 10) during the screening stage, the least influential variables were excluded from the subsequent RSM analysis in order to reduce model complexity and focus on the most relevant physicochemical drivers. Accordingly, kdeg(soil), Kow, and Pvap were retained as clearly significant factors. In addition, Ksw, Sol, and kdeg(air) were also included in the RSM study, although they were slightly below the significance threshold, because their effects were close enough to the cutoff to suggest potential relevance in the presence of curvature or interaction effects.

Step 17: BOX BEHNKEN

Create a new sheet and name it as 'BOX BEHNKEN'. Import data from the 'PREPROCESS-v3' sheet. Navigate to *DOE > Response Surface > Box Behnken*. Follow the specified selections outlined below and click on 'Execute'.

DoE Box Behnken

Number of Center Points per Block

1

Number of Replicates

1

Number of Blocks

1

☐ Random Standard order

Excluded Columns

>>

>

<

<<

Included Columns

Col2 -- Pvap

Col3 -- Sol

Col4 -- Kow

Col5 -- Ksw

Col6 -- kdeg_air

Col7 -- kdeg_soil

Execute

Cancel

Hint:
The Box–Behnken design was selected because it efficiently estimates quadratic models while requiring fewer experimental runs compared to full factorial designs, and avoids extreme corner points, making it suitable for stable environmental modelling studies.

Step 18: PREPROCESS- v4: Import & Select

Create a new sheet and name it as ‘PREPROCESS-v4’. Import data from the ‘BOX BEHNKEN’ sheet. Navigate to *Data Transformation > Data Manipulation > Select Column(s)* and select columns according to the specified selections below.

Select Column(s)

Excluded Columns

Col2 -- Standard Order

Col3 -- Block Number

Col4 -- Replicate Number

Col5 -- Point Type

>>

>

<

<<

Included Columns

Col6 -- Pvap

Col7 -- Sol

Col8 -- Kow

Col9 -- Ksw

Col10 -- kdeg_air

Col11 -- kdeg_soil

Execute

Cancel

Step 19: RESULTS-RSM: Create Sheet & Paste Data

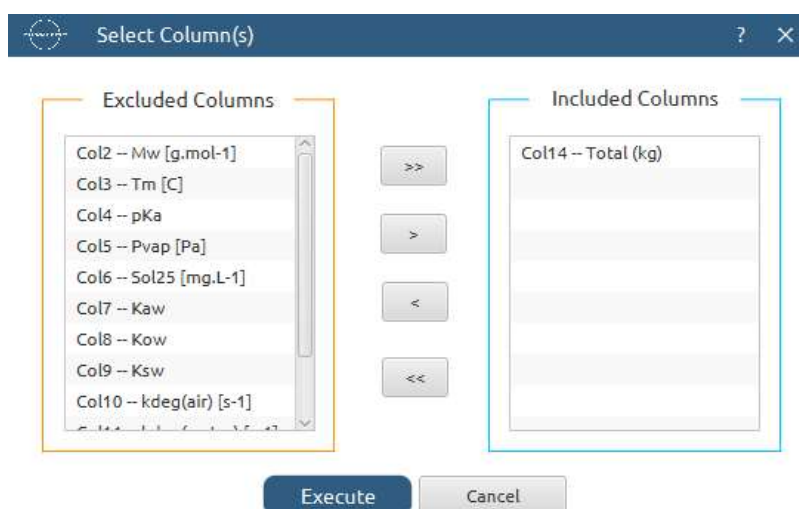
Create a new sheet and name it as 'RESULTS-RSM'. Copy and paste the table below on the input (left) spreadsheet.

Mw	Tm	pKa	Pvap	Sol	Kow	Kaw	Ksw	kdeg_air	kdeg_water	kdeg_sed	kdeg_soil	Total (kg)
500	195	0.5	1E-09	10	500050	0.0005	0.1	5.5E-07	5.5E-08	5.05E-10	5.05E-09	7.37001e+07
500	195	0.5	0.01	10	500050	0.0005	0.1	5.5E-07	5.5E-08	5.05E-10	5.05E-09	7.37655e+07
500	195	0.5	1E-09	10000	500050	0.0005	0.1	5.5E-07	5.5E-08	5.05E-10	5.05E-09	7.37001e+07
500	195	0.5	0.01	10000	500050	0.0005	0.1	5.5E-07	5.5E-08	5.05E-10	5.05E-09	7.37655e+07
500	195	0.5	1E-09	10	500050	0.0005	100	5.5E-07	5.5E-08	5.05E-10	5.05E-09	7.38901e+07
500	195	0.5	0.01	10	500050	0.0005	100	5.5E-07	5.5E-08	5.05E-10	5.05E-09	7.39547e+07
500	195	0.5	1E-09	10000	500050	0.0005	100	5.5E-07	5.5E-08	5.05E-10	5.05E-09	7.38901e+07
500	195	0.5	0.01	10000	500050	0.0005	100	5.5E-07	5.5E-08	5.05E-10	5.05E-09	7.39547e+07
500	195	0.5	0.005	10	100	0.0005	50.05	1E-07	5.5E-08	5.05E-10	5.05E-09	6.12784e+07
500	195	0.5	0.005	10000	100	0.0005	50.05	1E-07	5.5E-08	5.05E-10	5.05E-09	6.12784e+07
500	195	0.5	0.005	10	1000000	0.0005	50.05	1E-07	5.5E-08	5.05E-10	5.05E-09	8.08496e+07
500	195	0.5	0.005	10000	1000000	0.0005	50.05	1E-07	5.5E-08	5.05E-10	5.05E-09	8.08496e+07
500	195	0.5	0.005	10	100	0.0005	50.05	0.000001	5.5E-08	5.05E-10	5.05E-09	5.31330e+07
500	195	0.5	0.005	10000	100	0.0005	50.05	0.000001	5.5E-08	5.05E-10	5.05E-09	5.31330e+07
500	195	0.5	0.005	10	1000000	0.0005	50.05	0.000001	5.5E-08	5.05E-10	5.05E-09	7.31556e+07
500	195	0.5	0.005	10000	1000000	0.0005	50.05	0.000001	5.5E-08	5.05E-10	5.05E-09	7.31556e+07
500	195	0.5	0.005	5005	100	0.0005	0.1	5.5E-07	5.5E-08	5.05E-10	1E-10	3.53034e+07
500	195	0.5	0.005	5005	1000000	0.0005	0.1	5.5E-07	5.5E-08	5.05E-10	1E-10	3.18569e+09
500	195	0.5	0.005	5005	100	0.0005	100	5.5E-07	5.5E-08	5.05E-10	1E-10	3.00012e+08
500	195	0.5	0.005	5005	1000000	0.0005	100	5.5E-07	5.5E-08	5.05E-10	1E-10	3.18593e+09
500	195	0.5	0.005	5005	100	0.0005	0.1	5.5E-07	5.5E-08	5.05E-10	1E-08	1.99702e+07
500	195	0.5	0.005	5005	1000000	0.0005	0.1	5.5E-07	5.5E-08	5.05E-10	1E-08	3.84344e+07
500	195	0.5	0.005	5005	100	0.0005	100	5.5E-07	5.5E-08	5.05E-10	1E-08	3.51830e+07
500	195	0.5	0.005	5005	1000000	0.0005	100	5.5E-07	5.5E-08	5.05E-10	1E-08	3.86233e+07
500	195	0.5	1E-09	5005	500050	0.0005	0.1	1E-07	5.5E-08	5.05E-10	5.05E-09	8.06886e+07
500	195	0.5	0.01	5005	500050	0.0005	0.1	1E-07	5.5E-08	5.05E-10	5.05E-09	8.07525e+07
500	195	0.5	1E-09	5005	500050	0.0005	100	1E-07	5.5E-08	5.05E-10	5.05E-09	8.08734e+07
500	195	0.5	0.01	5005	500050	0.0005	100	1E-07	5.5E-08	5.05E-10	5.05E-09	8.09367e+07
500	195	0.5	1E-09	5005	500050	0.0005	0.1	0.000001	5.5E-08	5.05E-10	5.05E-09	7.29903e+07
500	195	0.5	0.01	5005	500050	0.0005	0.1	0.000001	5.5E-08	5.05E-10	5.05E-09	7.30557e+07
500	195	0.5	1E-09	5005	500050	0.0005	100	0.000001	5.5E-08	5.05E-10	5.05E-09	7.31807e+07
500	195	0.5	0.01	5005	500050	0.0005	100	0.000001	5.5E-08	5.05E-10	5.05E-09	7.32455e+07
500	195	0.5	0.005	10	500050	0.0005	50.05	1E-07	5.5E-08	5.05E-10	1E-10	3.16038e+09
500	195	0.5	0.005	10000	500050	0.0005	50.05	1E-07	5.5E-08	5.05E-10	1E-10	3.16038e+09
500	195	0.5	0.005	10	500050	0.0005	50.05	0.000001	5.5E-08	5.05E-10	1E-10	3.15254e+09
500	195	0.5	0.005	10000	500050	0.0005	50.05	0.000001	5.5E-08	5.05E-10	1E-10	3.15254e+09
500	195	0.5	0.005	10	500050	0.0005	50.05	1E-07	5.5E-08	5.05E-10	1E-08	4.54982e+07
500	195	0.5	0.005	10000	500050	0.0005	50.05	1E-07	5.5E-08	5.05E-10	1E-08	4.54982e+07
500	195	0.5	0.005	10	500050	0.0005	50.05	0.000001	5.5E-08	5.05E-10	1E-08	3.78057e+07
500	195	0.5	0.005	10000	500050	0.0005	50.05	0.000001	5.5E-08	5.05E-10	1E-08	3.78057e+07
500	195	0.5	1E-09	5005	100	0.0005	50.05	5.5E-07	5.5E-08	5.05E-10	1E-10	1.55376e+08
500	195	0.5	0.01	5005	100	0.0005	50.05	5.5E-07	5.5E-08	5.05E-10	1E-10	1.74744e+08

500	195	0.5	1E-09	5005	1000000	0.0005	50.05	5.5E-07	5.5E-08	5.05E-10	1E-10	3.12269e+09
500	195	0.5	0.01	5005	1000000	0.0005	50.05	5.5E-07	5.5E-08	5.05E-10	1E-10	3.18736e+09
500	195	0.5	1E-09	5005	100	0.0005	50.05	5.5E-07	5.5E-08	5.05E-10	1E-08	3.24302e+07
500	195	0.5	0.01	5005	100	0.0005	50.05	5.5E-07	5.5E-08	5.05E-10	1E-08	3.27370e+07
500	195	0.5	1E-09	5005	1000000	0.0005	50.05	5.5E-07	5.5E-08	5.05E-10	1E-08	3.85112e+07
500	195	0.5	0.01	5005	1000000	0.0005	50.05	5.5E-07	5.5E-08	5.05E-10	1E-08	3.85193e+07
500	195	0.5	0.005	5005	500050	0.0005	50.05	5.5E-07	5.5E-08	5.05E-10	5.05E-09	7.38488e+07

Step 20: Select Output Column

Navigate to *Data Transformation > Data Manipulation > Select Column(s)* and select only the column corresponding to the generated output, as shown below.



Step 21: PREPROCESS-v5: Merge Data

Create a new sheet and name it as 'PREPROCESS-v5'. Right-click on the input (left) spreadsheet and select 'Import from Multiple Spreadsheets'. Select the 'PREPROCESS-v4' as the Left Spreadsheet and 'RESULTS-RSM' as the Right Spreadsheet, then click on 'Execute' button.

	Col1	Col2	Col3	Col4	Col5	Col6
User Header	User Row ID					
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						

Show Spreadsheet Toolbar
 Import from File
 Import from Spreadsheet
 Import from Multiple Spreadsheets
 Adjust Spreadsheet Precision
 Export Spreadsheet Data
 Clear Spreadsheet

Join Configuration Steps

Step 1: PREPROCESS-v4 RESULTS-RSM

Join Type

☒ Concatenation ☐ Left Join ☐ Right Join ☐ Inner Join ☐ Full Outer Join

Left Spreadsheet

PREPROCESS-v4

Right Spreadsheet

RESULTS-RSM

Join Column

☒ Common header name ☐ Different header names

Execute Cancel

Step 22: PREPROCESS-v5: Select Columns

Navigate to Data Transformation > Data Manipulation > Select Column(s) and select the columns shown below.

Select Column(s)

Excluded Columns

Included Columns

Col2 -- Pvpap
Col3 -- Sol
Col4 -- Kow
Col5 -- Ksw
Col6 -- kdeg_air
Col7 -- kdeg_soil
Col8 -- Total (kg)

Execute Cancel

Step 23: PARETO-RSM

Create a new sheet and name it as 'PARETO-RSM'. Right-click on the input (left) spreadsheet and select 'Import from Spreadsheet' and select the 'PREPROCESS-v5'.

Navigate to *DOE > Post DoE Analysis > Pareto Analysis* and select the configuration as shown below.

Pareto Analysis

?

×

Dependent Variable

Col8 -- Total (kg)

Analysis Type

Main Effects + Two-Factor...

Level Of Significance

0.05

Excluded Columns

>

<

Factors

Col2 -- Pvap
Col3 -- Sol
Col4 -- Kow
Col5 -- Ksw

Covariates

>

<

DOE type:

Factorial / Screening

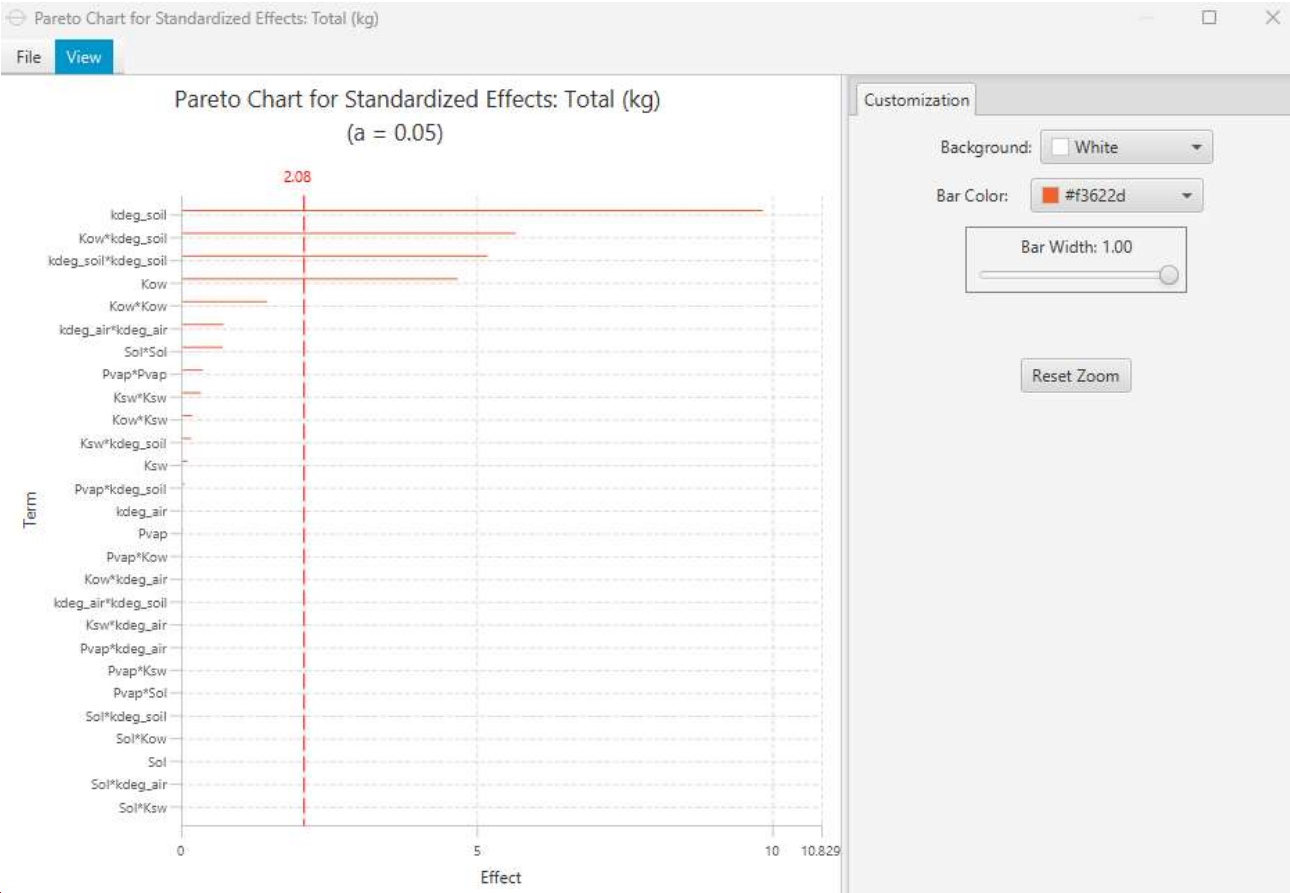
☒ Response Surface

☐ Include Center Points

Execute

Cancel

Output:



Observation:

The Pareto analysis of the Box Behnken results showed that $kdeg(soil)$, the interaction term $Kow \times kdeg(soil)$, the quadratic term $kdeg(soil)^2$, and Kow were the most important statistically significant contributors to the environmental fate response.

Step 24: Factorial-Plots-RSM

Create a new sheet and name it as 'FACTORIAL-PLOTS-RSM'. Right-click on the input (left) spreadsheet and select 'Import from Spreadsheet' and select the 'PREPROCESS-v5'.

Navigate to *DOE > Post DoE Analysis > Factorial Plot Analysis* and select the configuration as shown below.

Factorial Plot Analysis

Dependent Variable: Col8 -- Total (kg)

Analysis Type: Main Effects + Two-Factor...

Excluded Columns: (Empty)

Factors: Col2 -- Pvp, Col3 -- Sol, Col4 -- Kow, Col5 -- Ksw

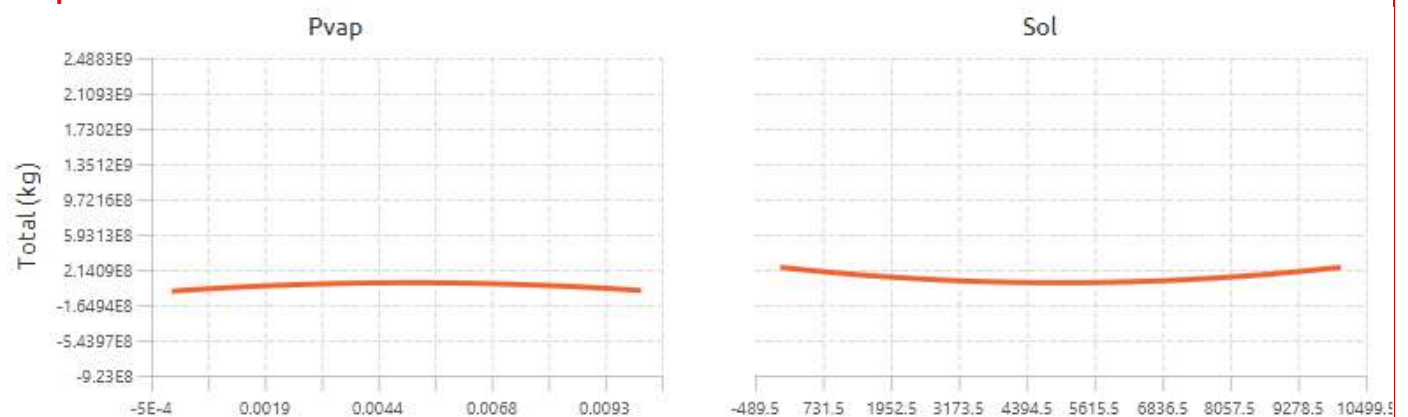
Specify Factor Values

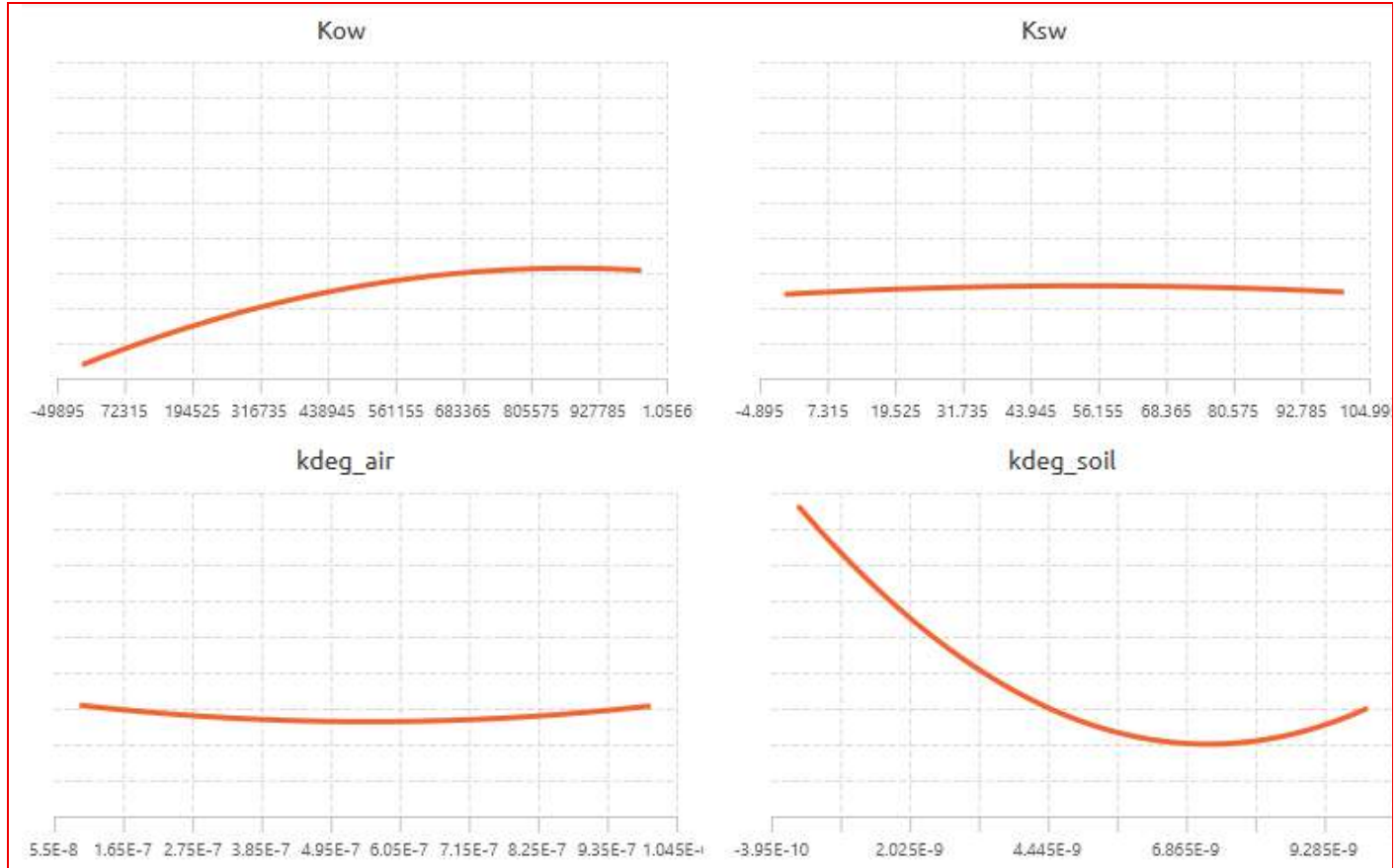
Covariates: (Empty)

DOE type: ☐ Factorial / Screening ☒ Response Surface

☐ Include Center Points

Execute Cancel

Output:

**Observation:**

The factorial plots obtained from the Box-Behnken design further highlight the presence of curvature effects.

Step 25: OPTIMIZATION

Create a new sheet and name it as 'OPTIMIZATION'. Right-click on the input (left) spreadsheet and select 'Import from Spreadsheet' and select the 'PREPROCESS-v5'.

Navigate to *DOE > Post DoE Analysis > Multi-Objective Optimization* and select the configuration as shown below.

Multi Objective Optimization

Analysis Type: Main Effects + Two-Factor...

Confidence Level: 95

Excluded Columns: [Empty Box]

Factors:

- Col2 – Pvap
- Col3 – Sol
- Col4 – Kow
- Col5 – Ksw
- Col6 – kdeg_air
- Col7 – kdeg_soil

Specify Factor Values

Dependent Variables:

Col8 – Total (kg) Target 1E+7 Scale: 1 Importance: 1

Specify Dependent Variable Ranges

☐ Optimize on the Discretized Space (Only factor values that appear in the dataset will be considered)

Execute Cancel

Output:

User Row ID							
	Optimal Settings - Solution						
	Pvap	Sol	Kow	Ksw	kdeg_air	kdeg_soil	
	0.0015009	1957.0324061	579703.6465934	59.6863679	5E-7	0E-7	
	Response Prediction						
	Response	Predicted Value	Standard Error	(95.0%) CI Lower	(95.0%) CI Upper	(95.0%) PI Lower	(95.0%) PI Upper
	Total (kg)	10000000.0000001	351317420.5930987	-720604571.7594274	740604571.7594277	-1311298072.0422013	1331298072.0422013
	Desirabilities						
	Total (kg)	Overall					
	1.0	1.0					

Observation:

Following the Box–Behnken design, a multi-objective optimization was performed with the target of minimizing the total environmental mass to approximately 10,000,000 kg. The optimal solution suggests that low degradation in soil ($kdeg(soil) = 1.0 \times 10^{-8} s^{-1}$) combined with relatively high Kow and moderate Pvap and Sol values leads to the desired outcome. The predicted total mass is close to the target, with an overall desirability of 1.0, indicating an excellent optimization result. However, the wide confidence and prediction intervals highlight a high level of uncertainty in the prediction, suggesting that while the model captures the trend, variability in the system remains significant and should be considered when interpreting the results.

These optimal conditions can be interpreted within an SSbD framework, as they suggest combinations of physicochemical properties that minimize environmental persistence and accumulation, thereby guiding the design of safer PFAS alternatives.

Final Isalos Workflow

Following the above-described steps, the final workflow on Isalos will look like this:

